A Simplified Model of Serpentine Locomotion

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Note. This document is an updated presentation of the dynamical modeling and control components of my 2017 MS thesis and paper [Racioppo and Ben-Tzvi, 2019].

Abstract

We revisit classical models of serpentine locomotion and show that the dynamics of snake robots admit a low-dimensional structure that is obscured by standard multibody formulations. While full models require solving constrained differential-algebraic systems, we demonstrate that, under mild assumptions, locomotion can be approximated by a single first-order equation governing the center-of-mass velocity. We identify the key mechanism underlying propulsion as a phase offset between lateral motion and body orientation, induced by anisotropic friction. This perspective yields (i) a simplified dynamical model that predicts steady-state velocity, (ii) a closed-form approximation for the optimal joint phase offset, and (iii) an interpretation of serpentine motion as a traveling-wave parameterization of a low-dimensional control manifold. Together, these results provide a unified and computationally efficient view of snake locomotion, connecting geometric gait design, physical modeling, and control.

1 Introduction

Serpentine locomotion, or lateral undulation, is the dominant mode of motion in biological snakes and has long served as a template for snake robot design. It consists of a traveling transverse wave that propagates along the body, generating thrust through interactions with the environment. Unlike standing waves, this motion relies on a coordinated phase relationship between body segments: each segment pushes laterally against the ground while the overall wave advances from head to tail. When coupled with anisotropic friction—higher resistance normal to the body than along it—this lateral motion is converted into forward propulsion.

In robotic implementations, this continuous body deformation is approximated by a finite set of rigid links connected by actuated joints. The standard parameterization of planar serpentine motion prescribes the joint angles as a traveling sinusoidal wave:

$$\theta_{\mathrm{ref},i} = \alpha \sin(\omega t + (i-1)\delta) + \phi_0, \tag{1}$$

for $i \in \{1, \dots, N-1\}$. Here, α is the amplitude of angular displacement, ω is the temporal frequency, and δ is the phase offset between adjacent joints. The parameter ϕ_0 induces a constant curvature and enables turning.

While this representation is typically viewed as a high-dimensional control input over $N-1$ joints, it in fact defines a low-dimensional structure: the entire body motion is generated by a small set of parameters $(\alpha, \omega, \delta, \phi_0)$ describing a traveling wave. In this sense, serpentine locomotion can be understood as motion constrained to a low-dimensional manifold embedded in the full configuration space of the robot.

Despite this inherent structure, most existing models of snake locomotion operate at the level of full multibody dynamics, requiring the solution of large systems of coupled differential-algebraic equations. While these models are physically accurate, they often obscure the underlying mechanisms that govern locomotion performance, such as the role of phase relationships and anisotropic drag.

In this work, we show that under mild assumptions, the dynamics of serpentine locomotion admit a significant reduction in complexity. In particular, we demonstrate that forward propulsion arises from a phase offset between lateral motion and body orientation, and that the resulting dynamics can be approximated

by a single first-order equation governing the center-of-mass velocity. This perspective enables a unified treatment of gait design, physical modeling, and control, and leads to (i) a simplified dynamical model that predicts steady-state speed, (ii) a closed-form approximation for the optimal phase offset, and (iii) an interpretation of serpentine motion as a traveling-wave parameterization of a low-dimensional control manifold.

By making this structure explicit, we aim to bridge the gap between detailed physical models and intuitive, interpretable descriptions of locomotion, providing tools that are both analytically tractable and physically grounded.

2 Related Work

Early work in snake robotics is largely attributed to Hirose [1993], who introduced the serpenoid curve as a mathematical description of serpentine locomotion. This formulation models the body of a snake as a traveling sinusoidal wave and remains the foundation for most subsequent work on snake robot kinematics.

Extensions of this idea have focused on alternative curvature parameterizations and their effect on locomotion efficiency. Ma [1999] analyzed different curvature profiles and compared their performance, while Chernousko [2005] proposed a complementary approach in which locomotion is constructed from elementary joint motions in low-degree-of-freedom systems. These works highlight that a wide range of locomotion patterns can be generated from relatively simple underlying structures.

A significant body of work has also focused on control and physical modeling. Liljebäck et al. [2012] and Liljebäck and Pettersen [2011] developed path-following and waypoint guidance controllers for snake robots executing serpenoid-like motions. In parallel, Liljebäck et al. [2010] established that anisotropic ground friction is a necessary condition for controllable serpentine locomotion, formalizing the role of directional frictional forces in generating propulsion. Further analysis of controllability and stability was provided by Liljebäck et al. [2011].

More comprehensive treatments of snake robot modeling and locomotion can be found in the survey by Transeth et al. [2009] and the monograph by Liljebäck et al. [2013], which cover both kinematic and dynamic approaches. Earlier experimental and control-oriented studies, such as Sato et al. [2002], also contributed to the development of practical locomotion strategies.

In addition to modeling and control, optimization of gait and design parameters has been widely studied, particularly with the goal of maximizing forward speed. This includes tuning link lengths, mass distributions, and control parameters to improve performance [Hirose, 1993, Chernousko, 2005].

Prior work has primarily approached serpentine locomotion from two complementary perspectives: kinematic descriptions based on traveling waves (e.g., the serpenoid curve), and high-fidelity dynamic models that capture frictional interactions and multibody effects. While both perspectives have led to significant advances, they often remain disconnected: kinematic models provide intuition but lack predictive power, whereas dynamic models are accurate but difficult to analyze and interpret.

This work aims to bridge this gap by making explicit the low-dimensional structure underlying serpentine locomotion. We show that the traveling-wave parameterization used in kinematic models induces a corresponding simplification in the dynamics, allowing the high-dimensional multibody system to be approximated by a single first-order equation governing center-of-mass motion. In this view, key quantities such as the phase offset and friction anisotropy emerge as the primary determinants of locomotion performance, providing a unified interpretation that connects gait design, physical modeling, and control.

3 Multibody Dynamic Model

We present a multibody dynamic (MBD) model of a snake robot, which serves as a high-fidelity reference for the simplified model developed in the previous section. Unlike reduced models that impose no-slip constraints, this formulation explicitly models frictional forces and internal constraint forces via Lagrange multipliers.

3.1 Kinematics and Constraints

The snake robot is modeled as N rigid links connected by revolute joints. Each link i has half-length L_i, mass m_i, and moment of inertia I_i. The generalized coordinates are:

$$q = [q_1, \ldots, q_N]^T, \quad q_i = [r_i, \phi_i]^T$$

where $r_i = [x_i, y_i]^T$ is the position of the center of mass and ϕ_i is the orientation.

Revolute joints impose $2(N-1)$ algebraic constraints, collected in:

$$\Phi(q, t) = 0.$$

The rotation matrix mapping global to local coordinates is:

$$R_i = \begin{bmatrix} \cos\phi_i & -\sin\phi_i \\ \sin\phi_i & \cos\phi_i \end{bmatrix}.$$

3.2 Friction Model

Friction is modeled using anisotropic coefficients c_n (normal) and c_t (tangential). The viscous friction force acting on link i is:

$$F_{f,i}(r_i, \phi_i) = -R_i^T \Omega_i R_i \dot{r}_i, \tag{2}$$

where $\Omega_i = \mathrm{diag}(c_{n,i}, c_{t,i})$.

Rotational friction is computed by approximating each half-link as n discrete masses with spacing:

$$l_{n,i} = \frac{2L_i}{2n+1}.$$

For angular velocity $\dot{\phi}_i$, segment velocities are:

$$v_{x_{n,i}} = \dot{\phi}_i l_{n,i} \cos\phi_i, \tag{3}$$

$$v_{y_{n,i}} = -\dot{\phi}_i l_{n,i} \sin\phi_i. \tag{4}$$

In the limit $n \to \infty$, this yields an equivalent force acting at:

$$l'_{\mathrm{eff},i} = \frac{1}{2}.$$

Defining:

$$v_{\mathrm{eff},i} = \begin{bmatrix} \dot{\phi}_i l'_{\mathrm{eff},i} \cos\phi_i \\ \dot{\phi}_i l'_{\mathrm{eff},i} \sin\phi_i \end{bmatrix},$$

the frictional torque is:

$$\tau_{f,i}(\phi_i) = l'_{\mathrm{eff},i} F_{f,i}(v_{\mathrm{eff},i}). \tag{5}$$

3.3 Equations of Motion

Let Q_A denote the generalized force vector and define:

$$M = \mathrm{diag}(m_1, m_1, I_1, \ldots, m_N, m_N, I_N).$$

The constrained variational equations are:

$$\delta q^T (M\ddot{q} - Q_A) = 0. \tag{6}$$

Introducing Lagrange multipliers λ, the system becomes the DAE:

$$\begin{bmatrix} M & \Phi_q^T & 0 \\ \Phi_q & 0 & 0 \\ 0 & 0 & I \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \\ \dot{q} \end{bmatrix} = \begin{bmatrix} Q_A \\ \gamma \\ \dot{q} \end{bmatrix}, \tag{7}$$

where Φ_q is the constraint Jacobian and γ contains acceleration-independent terms.

Initial conditions must satisfy:

$$\Phi_q \dot{q} = -\Phi_t.$$

The system state

$$Y = [\dot{q}^T \;\; \lambda^T \;\; q^T]^T$$

is obtained using a numerical DAE solver.

4 Single-Body Model of Lateral Undulation

We develop a simplified model of serpentine locomotion by approximating the dynamics of a multi-link snake robot with the averaged behavior of a single representative link. Under mild assumptions—low acceleration, small amplitude, and near-optimal phase offset—the high-dimensional system admits a reduced description that captures the dominant mechanisms governing propulsion.

4.1 Setup and Friction Model

Consider a snake with N links of uniform length L and mass m undergoing planar motion with link centers $r_i = [x_i \;\; y_i]^T$. Assuming motion primarily in the y-direction, the acceleration of link i under anisotropic viscous friction is:

$$\ddot{r}_i = -\frac{1}{m} R_i^T \Omega_i R_i \dot{r}_i \tag{8}$$

where $\Omega_i = \mathrm{diag}(c_{n,i}, c_{t,i})$ and R_i is the rotation matrix associated with orientation $\phi_i = a \sin(\omega t + (i-1)\delta)$.

Expanding the friction term:

$$-R_i^T \Omega_i R_i = -\begin{bmatrix} c_n \cos^2 \phi_i + c_t \sin^2 \phi_i & (c_n - c_t) \sin \phi_i \cos \phi_i \\ (c_n - c_t) \sin \phi_i \cos \phi_i & c_n \sin^2 \phi_i + c_t \cos^2 \phi_i \end{bmatrix}$$

The diagonal terms correspond to drag, while the off-diagonal terms generate propulsion through coupling between lateral velocity and orientation.

4.2 Assumptions

We consider the regime:

- Low period-averaged acceleration (near steady-state motion)
- Small amplitude $a \ll 1$
- Phase offset $\delta \approx \bar{\delta}$ (near optimal)

These assumptions enable linearization of trigonometric terms and justify time-averaging over an oscillation period.

The sinusoidal form of ϕ_i is not derived from the dynamics, but imposed by the joint control law, which enforces a traveling-wave gait. The analysis that follows should therefore be interpreted as computing the forces induced by a prescribed kinematic pattern, rather than solving for the motion from first principles. The reduced model is obtained by projecting the full multibody dynamics onto this low-dimensional manifold and averaging over an oscillation period. This approximation is valid when the motion is sufficiently smooth and coherent across the body, such that the behavior of a representative link captures the dominant dynamics of the system.

4.3 Kinematics of a Representative Link

Let link k satisfy $\phi_k = 0$ at $t = 0$, so that:

$$\phi_k = a\sin(\omega t)$$

The maximum lateral displacement is obtained by summing projections of neighboring links, yielding $m = \pi/\delta$. The lateral displacement is:

$$x_k \approx -La\sum_{i=1}^{\pi/\delta}\sin(\omega t + i\delta)$$

Differentiating:

$$\dot{x}_k \approx \frac{La\omega}{2}\csc\frac{\delta}{2}\sin\left(\omega t + \frac{\delta}{2}\right) \tag{9}$$

4.4 Propulsive Force Generation

The y-component of the friction force is:

$$F_{ky} = -\left[(c_n - c_t)\sin\phi_k\cos\phi_k\dot{x}_k + (c_n\sin^2\phi_k + c_t\cos^2\phi_k)\dot{y}_k\right]$$

The first term produces propulsion. Using small-angle approximations:

$$F_{ky,\text{prop}} \approx -\frac{La^2\omega(c_n - c_t)}{2}\csc\frac{\delta}{2}\sin(\omega t)\sin\left(\omega t + \frac{\delta}{2}\right)$$

Averaging over one period and normalizing:

$$\bar{F}_p(N) \approx J_1(2)^2\frac{(a\omega)^2(c_n - c_t)}{4\pi N}\cot\left(\frac{2.05}{N} + \frac{1.08}{N^2}\right) \tag{10}$$

This shows propulsion increases monotonically with N.

4.5 Angular Dynamics and Control

The angular dynamics of link k are:

$$I\ddot{\phi} = \tau_u - f_v\dot{\phi} + f_i(\phi, \dot{\phi})$$

with

$$f_v = \frac{2}{3}c_n\left(\frac{L}{2}\right)^2$$

Using PD control:

$$\tau_u = k_p[\alpha\sin(\omega t) - \phi] + k_d[\alpha\omega\cos(\omega t) - \dot{\phi}]$$

Assuming cancellation of nonlinear terms (partial feedback linearization):

$$I\ddot{\phi} + c\dot{\phi} + b\phi = k_p\alpha\sin(\omega t) + k_d\alpha\omega\cos(\omega t)$$

The steady-state amplitude is:

$$\bar{a} = \alpha\omega\frac{kdI\omega^2 - bkd + bc}{(I\omega^2 - b)^2 + c^2\omega^2} \tag{11}$$

4.6 Reduction to a Single Dynamical Equation

Combining the above, the center-of-mass velocity $Y = \dot{y}$ is approximated by:

$$\dot{Y}(t) = f_x(t) - f_y(t)Y(t) \tag{12}$$

where

$$\begin{cases} f_x(t) = C_1 \bar{a}\omega \sin\bar{\phi}\cos\bar{\phi}\sin\left(\omega t + \frac{\bar{\delta}}{2}\right) \\ f_y(t) = C_2\left(c_n \sin^2\bar{\phi} + c_t\cos^2\bar{\phi}\right) \end{cases} \tag{13}$$

Under the stated assumptions, the $3N$-dimensional dynamics reduce to a single first-order equation governing forward velocity. Propulsion arises from the phase interaction encoded in $f_x(t)$, while $f_y(t)$ captures orientation-dependent drag.

The model reveals that forward motion is governed by the phase offset between lateral velocity and orientation. When these signals are in phase, motion is reciprocal and produces no net displacement. A phase shift, together with anisotropic friction, produces net propulsion.

5 Optimal Phase Offset and Traveling-Wave Structure

In this section, we analyze the structure of the phase-offset optimization problem introduced in Liljebäck et al. [2013]. Starting from the original double-sum formulation, we show that it can be reduced to a single weighted sum over phase differences along the body.

This reduction reveals that the objective depends only on relative phase differences, with weights that vary systematically along the chain. As a consequence, maximizing propulsion corresponds to distributing phase differences approximately uniformly across the body, leading to the approximation

$$\delta_{\text{opt}} \approx \frac{\pi}{N-1}.$$

This provides a geometric interpretation of the optimal gait: the body configuration approximates a single spatial period of a traveling wave.

5.1 Background and Problem Formulation

Following Liljebäck et al. [2013], the phase offset that maximizes forward velocity can be approximated as the maximizer of:

$$k_\delta = \sum_{i=1}^{N-1}\sum_{j=1}^{N-1} c_{ij}\sin((j-i)\delta) \tag{14}$$

where $C = AD^T(DD^T)^{-1}$, and

$$A = \begin{bmatrix} 1 & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & -1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 1 & -1 \end{bmatrix}.$$

This expression encodes how phase differences between joints contribute to propulsion. The original expression is written as a double sum over all joint pairs. We now show that, due to the structure of the matrix C, this expression can be rewritten as a single sum indexed by phase differences.

5.2 Reduction to a Single Summation

We now derive a closed-form approximation by analyzing the structure of C.

Let $n = N - 1$ and define $B = (DD^T)^{-1}$. Since DD^T is symmetric, B is also symmetric.

Proposition 1. Let $B = (DD^T)^{-1} \in \mathbb{R}^{n\times n}$. The sum of the k-th diagonal of B is

$$b_k = \frac{T_k}{n+1}, \quad \text{where} \quad T_k = \frac{k(k+1)(k+2)}{6}.$$

Here, the k-th diagonal is indexed from the outermost diagonal (bottom-left / top-right) inward toward the main diagonal.

Proof. We establish three properties of B.

(1.1) The entries along the boundary of B increase linearly in steps of $\frac{1}{n+1}$.

(1.2) The first n diagonals of B are recursively generated from the bottom row and scale with $\frac{1}{n+1}$.

(1.3) The sum of the main diagonal is $b_n = \frac{T_n}{n+1}$.

Proof of (1.1). Consider the first column of B. The defining relation $DD^TB = I$ implies:

$$-b_{n,1} + 2b_{n-1,1} - b_{n-2,1} = 0 \quad \Rightarrow \quad b_{n-1,1} = 2b_{n,1},$$

$$-b_{n-2,1} + 2b_{n-1,1} - b_{n-3,1} = 0 \quad \Rightarrow \quad b_{n-2,1} = 3b_{n,1}.$$

Proceeding inductively, suppose:

$$b_{n-i,1} = (i+1)b_{n,1}, \quad i \geq 2.$$

Then:

$$-b_{n-(i+1),1} + 2b_{n-i,1} - b_{n-(i-1),1} = 0$$

implies:

$$b_{n-(i+1),1} = 2(i+1)b_{n,1} - ib_{n,1} = (i+2)b_{n,1}.$$

Thus, the first column increases linearly. At the top:

$$b_{1,1} = n\, b_{n,1}.$$

Since $b_{n,1} = \frac{1}{n+1}$, we obtain:

$$b_{1,1} = \frac{n}{n+1}.$$

By symmetry, the same structure holds for the last column, and therefore for the top and bottom rows.

Proof of (1.2). Consider the i-th element of the k-th diagonal:

$$b_{n-k-(i-1),\, i+k-1}, \quad k \in [1,n],\ i \in [1,k].$$

These entries satisfy the recurrence:

$$-b_{n-k-(i-1),\, i+k-1} + 2b_{n-k-i,\, i+k} - b_{n-k-(i+1),\, i+k+1} = 0,$$

with boundary condition:

$$b_{n-k-(i-1),\, i+k-1} = 2b_{n-k-i,\, i+k}$$

at the second-to-bottom row.

Thus, each diagonal is generated recursively from the bottom row, and all diagonals share the same structure up to the scaling factor $\frac{1}{n+1}$.

Proof of (1.3). The matrix

$$DD^T = \begin{bmatrix} 2 & -1 & & 0 \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ 0 & & -1 & 2 \end{bmatrix}$$

is the second-difference matrix.

Using $DD^T B = I$ and taking traces:

$$\mathrm{tr}(DD^T B) = n.$$

This yields the identity:

$$\sum_{i=1}^{n-1} (b_{i,i} - b_{i,i+1}) + b_{n,n} = 1.$$

Let $f(k)$ denote the sum of the k-th diagonal (counted from the outside inward). Then the recursion:

$$f_n = 2f_{n-1} - f_{n-2} + \frac{n(n-1)}{2}$$

matches the tetrahedral sequence:

$$f_n = \frac{T_n}{n+1}.$$

Conclusion. By induction on n, all diagonal sums satisfy:

$$b_k = \frac{T_k}{n+1},$$

which proves Proposition 1.

□

Proposition 2. The sum of the k-th diagonal of $C = AD^T(DD^T)^{-1}$ is:

$$c_k = \pm\frac{2t_k}{n+1}, \quad t_k = \frac{k(k+1)}{2},$$

where the sign is positive in the upper triangle and negative in the lower triangle.

Proof. Multiplication by D^T subtracts adjacent diagonals:

$$\sum_i b_{i,k+i} \mapsto \sum_i (b_{i,k+i} - b_{i,k+i-1}).$$

Thus, diagonal sums become:

$$t_k - t_{k-1} = t_k,$$

where t_k are triangular numbers.

Multiplication by A adds adjacent rows. Equivalently, each diagonal is added to the one above it, doubling its contribution (except for boundary corrections).

Accounting for boundary terms, we obtain:

$$c_k = \pm\frac{2t_k}{n+1}.$$

□

Proposition 3. The objective function simplifies to:

$$k_\delta = \sum_{k=1}^{N-1} \frac{2k(k+1)}{N} \sin((N-1-k)\delta). \tag{15}$$

Proof. Since each diagonal corresponds to $j - i = k$, the sine term is constant along each diagonal. Using the result of Proposition 2:

$$k_\delta = \sum_{k=1}^{N-1} \left(\frac{2t_k}{n+1} \sin(k\delta) - \frac{2t_k}{n+1} \sin(-k\delta) \right).$$

Using $\sin(-x) = -\sin x$:

$$k_\delta = \frac{2}{n+1} \sum_{k=1}^{N-1} t_k \sin(k\delta) + \frac{2}{n+1} \sum_{k=1}^{N-1} t_k \sin(k\delta).$$

Substituting $t_k = k(k+1)/2$ and $n = N - 1$ yields:

$$k_\delta = \sum_{k=1}^{N-1} \frac{2k(k+1)}{N} \sin((N-1-k)\delta).$$

□

This reduction shows that the objective depends only on phase differences between links, with weights that increase along the body. The contribution of each phase difference is therefore not uniform, but follows a structured weighting induced by the underlying model.

5.3 Closed-Form Approximation

Expanding and differentiating:

$$1 + \frac{N(N-1)}{N+1} \cos(\bar{\delta}) + \frac{N-2}{N+1} \cos(\bar{\delta} N) = \cos(\bar{\delta}(N-1))^{N-2}$$

Taylor expansion yields:

$$\delta_{\text{opt}} \approx \frac{3\pi}{N} - \frac{3\pi}{N^2} \tag{16}$$

and

$$\delta_{\text{opt}} \approx \frac{2\pi}{N} + \frac{2\pi}{N^2} - \frac{4\pi}{3N^3} + \frac{4\pi}{3N^4} \tag{17}$$

An alternative approximation is:

$$\delta_{\text{opt}} \approx \frac{4.10}{N} + \frac{2.15}{N^2} \tag{18}$$

5.4 Interpretation

A simpler but fairly accurate approximation for the above is:

$$\delta_{\text{opt}} \approx \frac{\pi}{N-1} \tag{19}$$

This leads to a simple geometric interpretation: the optimal phase offset corresponds to distributing a total phase difference of approximately π across the body. In this configuration, the first and last joints are approximately π out of phase, and the body approximates a single spatial period of a traveling wave.

6 Partial Feedback Linearization

An N-link snake robot moving in the plane can be modeled using a multibody dynamic approach as a serial chain of straight links joined by revolute joints. The system's dynamics are described by $3N$ differential algebraic equations (DAEs) for the links, complemented by $2(N-1)$ algebraic constraint equations for the revolute joints.

The generalized coordinates for the i-th link's center of mass (CM) are specified by a vector $q_i = [r_i, \phi_i]^T$, where $r_i = [x_i, y_i]^T$ denotes the global position in the xy-plane, and ϕ_i is its global orientation. Introducing a vector of Lagrange multipliers $\lambda_i = [\lambda_{x,i}, \lambda_{y,i}]^T$ to enforce the joint constraints, a diagonal 3×3 generalized mass matrix M_i, a generalized external force vector $F_{\text{ext},i}$, and a vector of generalized actuation forces U_i for the i-th link, the system's equations of motion can be written as:

$$\begin{cases} M_i \ddot{q}_i = F_{\text{ext},i} + f_i(q, \dot{q}) + \lambda_i + U_i \\ \Phi_j(q, t) = 0 \end{cases} \tag{20}$$

Here, $f_i(q, \dot{q})$ represents internal forces and torques (e.g., Coriolis and centrifugal terms), while $\Phi_j(q, t) = 0$ represents the j-th revolute joint constraint equation. Since joint torques are the only control inputs, $U_i = [0, 0, \tau_i]^T$.

The goal of feedback linearization is to transform a nonlinear system into a linear one by canceling nonlinear terms with appropriate control inputs. For a snake robot, the angular dynamics are of primary interest. As observed in [**?**], the control input is decomposed as:

$$U_i = U_{\text{lin},i} + \bar{U}_i \tag{21}$$

The term $\bar{U}_i$ tracks a desired angular trajectory, while $U_{\text{lin},i}$ cancels nonlinear dynamics.

The full system can be written as:

$$M\ddot{q} = F_{\text{ext}}(q, \dot{q}) + F_{\text{int}}(q, \dot{q}) + \Phi_q^T \lambda + U \tag{22}$$

The angular component for link i is:

$$I_i \ddot{\phi}_i = (F_{\text{ext},i})_\phi + (F_{\text{int},i})_\phi + (\Phi_q^T \lambda)_i + \tau_i \tag{23}$$

Setting:

$$\tau_i = -(F_{\text{ext},i})_\phi - (F_{\text{int},i})_\phi - (\Phi_q^T \lambda)_i + \bar{U}_i$$

yields:

$$I_i \ddot{\phi}_i = \bar{U}_i \tag{24}$$

The DAE system is:

$$\begin{bmatrix} M & \Phi_q^T \\ \Phi_q & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} Q_A \\ \gamma \end{bmatrix} \tag{25}$$

Expanding:

$$M\ddot{q} + \Phi_q^T \lambda = Q_A \tag{26}$$

$$\Phi_q \ddot{q} = \gamma \tag{27}$$

Solving:

$$\ddot{q} = M^{-1}(Q_A - \Phi_q^T \lambda) \tag{28}$$

$$\Phi_q M^{-1}(Q_A - \Phi_q^T \lambda) = \gamma \tag{29}$$

$$\Phi_q M^{-1} Q_A - \Phi_q M^{-1} \Phi_q^T \lambda = \gamma \tag{30}$$

Thus:

$$\Phi_q M^{-1} \Phi_q^T \lambda = \Phi_q M^{-1} Q_A - \gamma \tag{31}$$

Let $A = \Phi_q M^{-1} \Phi_q^T$. Then:

$$\lambda = A^{-1}(\Phi_q M^{-1} Q_A - \gamma) \tag{32}$$

Substituting λ back yields the linearizing control.

A key distinction of this formulation is that the feedback linearization is performed directly within the Differential-Algebraic framework, rather than on a reduced-order ODE system. In standard robotics literature, it is common practice to eliminate algebraic constraints via coordinate reduction before applying feedback linearization. However, by maintaining the redundant coordinate set and explicitly solving for the Lagrange multipliers λ, the controller gains direct access to the internal constraint forces. This allows for a 'Descriptor-based Linearization' that avoids the symbolic explosion associated with the joint-space Mass matrix of multi-link systems. By canceling the components of $(\Phi_q^T \lambda)$ within the control law, the actuator effectively 'transparentizes' the joint, decoupling the link's angular acceleration from the complex chain dynamics. This approach leverages the structured sparsity of the DAE matrices to maintain computational efficiency while providing a mathematically exact linearization of the internal couplings.

7 Path Following

A snake robot driving its joint angles according to Equation (8) moves in a straight line when $\phi_0 = 0$ and executes a turn with constant curvature for constant ϕ_0 [Sato et al., 2002]. To enable path following, ϕ_0 is dynamically adjusted.

A common approach [Liljebäck et al., 2013] defines:

$$\phi_{0,\text{ref}} = A_h(h_{\text{ref}} - h) \tag{33}$$

where:

$$h = \frac{1}{N-1} \sum_{i=1}^{N-1} \theta_i$$

The desired heading is:

$$h_{\text{ref}} = \arctan\left(\frac{P_Y}{\Delta - P_X}\right) \tag{34}$$

An alternative cross-track error:

$$P_{X,i,j} = \sqrt{(K_{i,j,x} - E_{j,x})^2 + (K_{i,j,y} - E_{j,y})^2} \tag{35}$$

Signed error:

$$P_{X,j} = -\text{sgn}(\theta_{\text{rel}} - \theta_{\text{wp}}) P_{X,j}$$

with:

$$\theta_{\text{rel}} = \arctan 2(v_{i,j,y}, v_{i,j,x}), \quad \theta_{\text{wp}} = \arctan 2(w_{j,y}, w_{j,x}) \tag{36}$$

PID heading:

$$h_{\text{ref},j} = K_P P_{X,j} + K_I \int P_{X,j} dt + K_D \dot{P}_{X,j} \tag{37}$$

Further refinements include adaptive gains:

$$K_P = p_1(\arctan 2(p_2 \phi_{\text{AoA}}) + p_3) \tag{38}$$

8 Experiments

The reduced model was compared to a full multibody simulation. Simulations were performed using MATLAB `ode45` over a 30-second interval, with the system initialized from rest.

At fixed parameter settings, the reduced model reproduces key qualitative features of the multibody dynamics. In particular, both models exhibit:

- A transient oscillatory regime in the center-of-mass velocity
- Convergence to a steady-state forward speed
- A dependence of steady-state speed and oscillation amplitude on the driving frequency ω

Across a range of frequencies, the reduced model captures the characteristic relationship between oscillation amplitude and steady-state velocity, with higher frequencies producing lower oscillation amplitudes and higher steady-state speeds.

Quantitatively, the agreement depends on the operating regime. For sufficiently large ω, the model achieves:

- $< 1\%$ error in steady-state amplitude for $\omega > 10$ Hz

while for lower frequencies:

- $\sim 40\%$ error at $\omega = 2$ Hz

These discrepancies can be traced directly to the approximations used in the reduced model. In particular, the derivation assumes (i) small-amplitude motion, (ii) near steady-state dynamics, and (iii) a separation of time scales between the fast oscillatory motion and the slower evolution of the center-of-mass velocity.

At low frequencies, the time-scale separation assumption breaks down: the forward velocity $Y(t)$ no longer evolves slowly relative to the oscillatory terms in $f_x(t)$ and $f_y(t)$, and the averaging argument used to obtain the reduced model becomes invalid. As a result, transient oscillations interact with the forward dynamics, leading to larger discrepancies in both amplitude and steady-state speed.

Similarly, for larger amplitudes, the small-angle approximations used in deriving $\dot{x}_k$ and the propulsive force no longer hold. In this regime, higher-order geometric effects and coupling between links become significant, and the assumption that a single representative link captures the behavior of the full system is no longer accurate.

Finally, the reduced model collapses the full multibody dynamics into a small number of effective coefficients (e.g., C_1, C_2), which implicitly absorb the effects of inter-link coupling and constraint forces. While this approximation captures the dominant behavior in regimes where the motion is smooth and coherent, it becomes sensitive to parameter choices when these conditions are not met.

More broadly, the reduced model is best interpreted as capturing the dominant dynamical structure of serpentine locomotion, rather than serving as a uniformly accurate predictor across all parameter regimes. Within its regime of validity, it provides a compact description of how phase relationships and anisotropic friction give rise to propulsion. Outside this regime, it retains qualitative agreement—such as the dependence of steady-state speed on frequency and body length—but loses quantitative accuracy due to the breakdown of the underlying approximations.

9 Conclusion

We revisited classical models of serpentine locomotion and showed that, despite the apparent complexity of multibody dynamics, the system admits a simple underlying structure. In particular, we demonstrated that locomotion can be understood as a low-dimensional traveling-wave process, in which both kinematics and dynamics are governed by a small set of parameters.

By analyzing the interaction between body orientation, lateral motion, and anisotropic friction, we derived a reduced dynamical model in which forward velocity is governed by a single first-order equation. This reduction makes explicit the mechanism of propulsion as a phase offset between lateral velocity and body orientation.

We further showed that the optimal joint phase offset admits a closed-form approximation and corresponds geometrically to a configuration in which the body spans a single spatial period of a traveling wave. This provides a direct link between gait design and locomotion performance.

Together, these results unify kinematic descriptions, dynamic models, and control strategies under a common low-dimensional framework. Beyond computational efficiency, this perspective provides a more interpretable understanding of serpentine locomotion and may inform the design of more effective control policies and robot morphologies.

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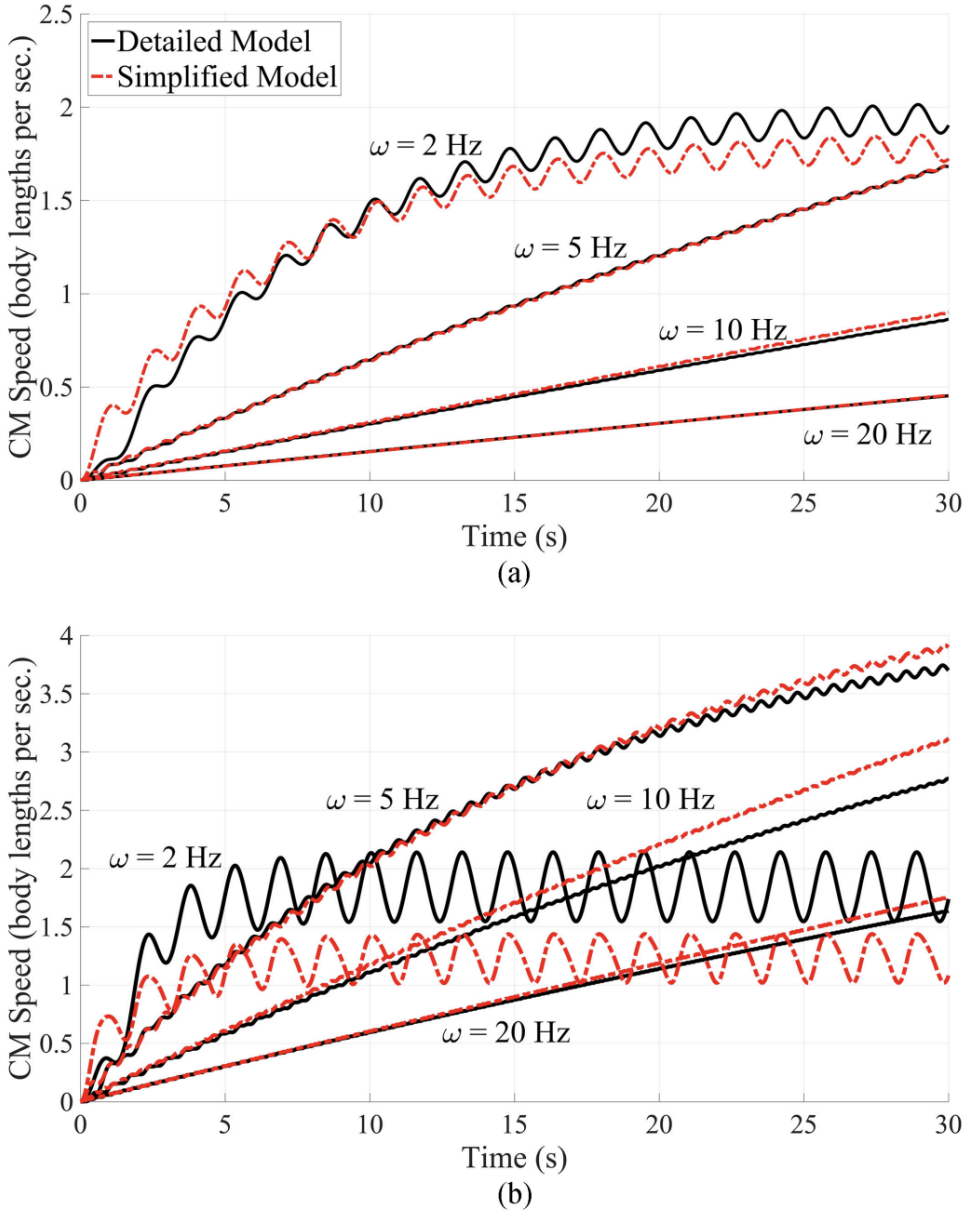

Figure 1: Comparison of the full multibody dynamic model (dark) with the single-body model (light) at various oscillation frequencies for (a) $\alpha = 0.5$ rad and (b) $\alpha = 1$ rad. The reduced model reproduces the characteristic transient oscillations and convergence to steady-state velocity.

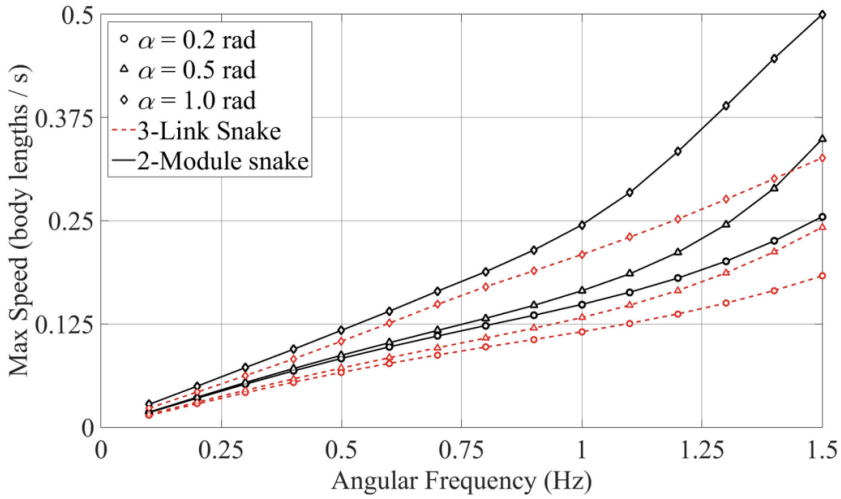

Figure 2: Simulated steady-state speeds for snake robots with different body lengths as the driving amplitude and frequency are varied. Increasing frequency leads to higher steady-state speeds, while longer bodies produce greater propulsion due to increased phase-coherent interactions.

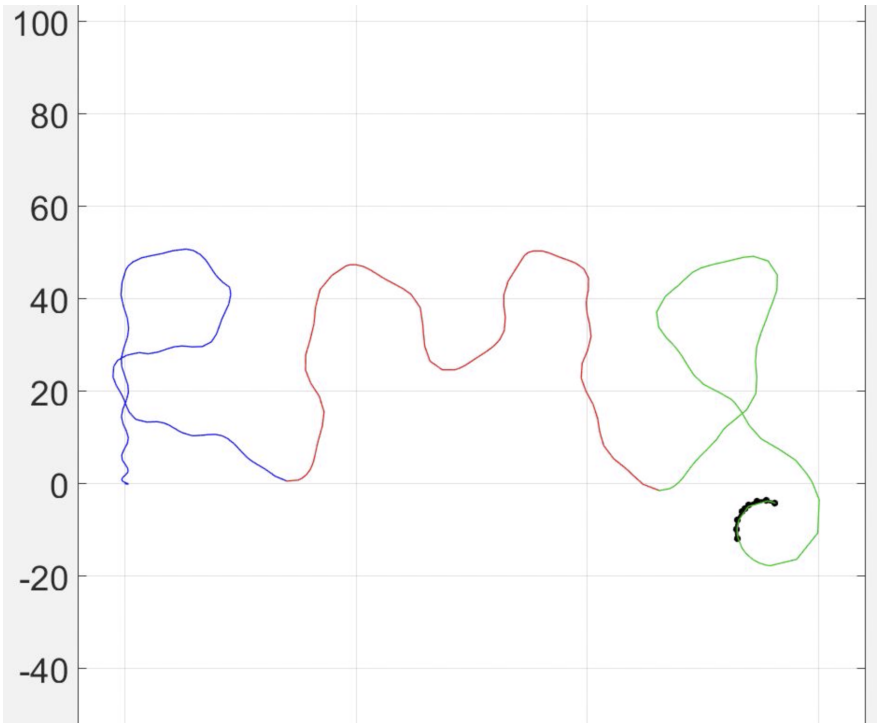

Figure 3: Simulated path-following control using waypoint guidance. The snake robot traces a sequence of waypoints to spell the letters "RML," demonstrating the effectiveness of the control strategy in generating complex planar trajectories.